

HANBEELMS TECHNICAL NOTEBOOK

The Complete Book of F1 RC Modelling & Real Car Engineering

Real Formula 1 architecture, how it translates into 1:10 scale RC modelling, and an honest reconciliation of where scale physics diverges from the real thing.



1. 🏎️ Real F1 Car Architecture

1.1 Aerodynamics & Bodywork

Downforce philosophy (2022–2025 "ground-effect" era)

- The current-generation F1 car generates the majority of its downforce from the **floor**, not the wings — a deliberate shift away from the wake-heavy, wing-dominant cars of 2017–2021 so cars can follow each other more closely.
- **Venturi tunnels**: two shaped channels running under the sidepods from the front of the floor to the diffuser. Air is accelerated through a narrowing throat, dropping pressure underneath the car (Bernoulli's principle) and sucking the floor toward the track — "ground effect."
- **Diffuser**: the upward-kicking ramp at the rear of the floor that expands the accelerated underfloor air back to ambient pressure gradually, sustaining the low-pressure effect for as long as possible without stalling the airflow.
- **Sidepod design philosophy** splits into two schools:
 - **Downwash/undercut** (e.g., Red Bull RB19): an aggressively scooped sidepod inlet that manages airflow over and around the sidepod to feed the floor edge and diffuser cleanly.
 - **"Zeropod"** (Mercedes W13/W14 concept): a narrow, minimal sidepod intended to maximize floor-edge airflow at the cost of internal cooling packaging — a real trade-off, not a free win.
- **Front/rear wings** are now a smaller share of total downforce than the floor, but still control airflow direction into the floor and over the rear.

2026 regulation shift — a real architecture change

- Venturi tunnels are **removed** for 2026: a flatter floor, smaller tunnels, reduced throat height, a less aggressive diffuser expansion.
- Peak downforce falls roughly **30%**, drag falls roughly **55%**, minimum weight drops from 800 kg to **768 kg**.
- New **active aerodynamics**: movable surfaces switching between a high-downforce "corner mode" and low-drag "straight-line mode" within a single lap.

1.2 Mechanical & Power Unit

Suspension geometry

- **Pushrod**: rod pushes up into the rocker as the wheel moves into bump.
- **Pullrod**: rod pulls down on the rocker as the wheel droops.

- Since 2022, this choice is as much an **aerodynamic** decision as a mechanical one — anti-dive/anti-squat geometry keeps the floor at its target ride height under braking/acceleration, protecting the ground-effect floor's performance.
- **Case study:** Ferrari popularised front pushrod suspension in the late 2010s, then returned to pullrod for the 2022 rules cycle — geometry follows the aero regulations of the era.

Hybrid power unit & ERS (through 2025)

- 1.6L turbocharged V6 ICE plus two recovery systems: **MGU-K** (kinetic, recovers braking energy) and **MGU-H** (heat, recovers turbo exhaust energy).

2026 power unit — a real architecture change

- **MGU-H deleted entirely**; turbocharged V6 ICE remains, revised.
- MGU-K deployment nearly triples: **120 kW → 350 kW (469 hp)** — close to a 50/50 electric/ICE power split.
- Recoverable energy per lap nearly doubles to **~8.5 MJ**, deployable in bursts up to **4 MJ** (~11.5s of full ERS-K power), and multiple bursts per lap are now allowed if the battery can sustain them.
- **Engineering implication:** with MGU-H gone, turbo-lag management and battery thermal/energy strategy replace exhaust-energy harvesting as the dominant electrical challenge.

Brakes

- Carbon-carbon disc and pad material for extreme heat tolerance under repeated high-deceleration events.
- **Brake-by-wire** on the rear axle blends friction braking with MGU-K regeneration so pedal feel stays consistent regardless of how much the electric motor recovers at that instant.

1.3 Team/Car Case Study: Red Bull RB19 (2023)

- **Livery:** navy-blue base with red accents, evolved subtly from the RB18 — same core layout and matte finish, expanded white sidepod highlights for the new title partner (Oracle).
- **Sidepod evolution mid-season:** at Azerbaijan, the sidepod inlet aperture was widened while reducing its height, improving undercut airflow — an even more aggressive version followed at Hungary. In-season iteration, not a fixed launch-spec design.
- **Front axle line** moved slightly forward vs. the RB18, changing how the front wing's wake interacts with the front wheels and how far that wake travels before reaching the floor and sidepods.
- Characterised overall as an **evolution**, not a revolution, of the 2022 ground-effect concept.

2. RC Modelling Translation

2.1 Chassis Selection & Scale

- **Scale:** 1:10 dominates dedicated F1-replica RC racing (shared parts/tire ecosystem with 1:10 touring/pan-car racing). 1:12 exists for smaller study builds but is less common for club racing.
- **Reference chassis — Tamiya F104 / F104 Pro II:**
 - Double-deck FRP frame: **2.0 mm** upper deck, **3.2 mm** lower deck.
 - **Longitudinal battery placement** down the centreline — a scale echo of how a real F1 car's heaviest single mass is packaged low and central for polar moment of inertia.
 - Front independent coil-spring suspension; rear **1.5 mm FRP T-bar** with friction plates and a damper — simplified, low-maintenance layout appropriate for a scale-appearance-first chassis.
 - **F104W** reuses F103-generation front suspension arms specifically to replicate early-1990s F1 track-width proportions.
 - **F104X1** adds a roll-centre damper and an X-shaped upper deck for stiffness — the RC equivalent of a stiffer monocoque.
 - 3D-printed custom parts (nose cones, mirror stalks, diffuser fins) are common enthusiast additions on top of the kit's injection-moulded body.

2.2 Translating Real Tech to RC

Real F1 concept	RC translation	Honest limitation
Venturi floor tunnels + diffuser	Scale plastic/resin floor with cosmetic tunnel and diffuser detail	Purely cosmetic on most kits — real ground-effect suction isn't reproducible without a sealed, purpose-engineered underbody
Front/rear wings	Scale ABS/polycarbonate wings, fixed or adjustable angle	Can produce some real aero load at speed, but nowhere near proportionally to a real car
Pushrod/pullrod inboard suspension	Bellcrank/rocker-actuated coil-overs inside the chassis rails	Kept mainly for scale appearance, not for the aero benefit real F1 gets from hiding suspension from airflow
Carbon-carbon brakes + brake-by-wire blending	Single brushless motor providing drive and regen-style braking via ESC throttle/brake mixing	No separate friction brake system exists — "braking" is entirely motor braking commanded through the ESC

Real F1 concept	RC translation	Honest limitation
MGU-K electric deployment	N/A — the RC car is already fully electric	The challenge inverts: RC engineers manage heat and current draw, not "adding" electric assist to combustion

2.3 Electronics & Power

Motor

Brushless is standard beyond beginner kits: higher efficiency, no brush/commutator wear, far higher achievable RPM per size/weight. Output is rated in **Kv** (RPM per volt, unloaded): a 3000 Kv motor on 2S LiPo (8.4V full) spins **~25,200 RPM** unloaded. Higher Kv trades low-end torque for top-end RPM — mirroring how a real F1 ICE is high-revving and relatively low-torque, reliant on gearing.

ESC (Electronic Speed Controller)

- Rated for the pack's cell count: **2S = 7.4V**, **3S = 11.1V** are the common on-road range.
- Current rating should sit roughly **10% above** the motor's burst-current rating.
- **Low-voltage cutoff (LVC)** configured at ~3.0V per cell — protects the LiPo from over-discharge, the RC equivalent of protecting a real high-voltage battery.

Servo (steering)

- 1:10 on-road racing typically targets **15–20 kg·cm** torque at a low-profile (~30mm or shorter) form factor.
- Representative part: a coreless digital servo (e.g. Futaba FT5478M) delivers ~9.2 kg·cm torque and 0.07 sec/60° speed at 7.4V, 40.1×20.0×28.5mm — speed matters as much as torque for close-quarters pan-car racing.

Battery placement

LiPo packs (2S most common) mount **low and central**, usually longitudinally along the chassis centreline — the single largest mass in the car, dominating centre-of-gravity height and yaw polar moment of inertia, exactly what a real F1 team obsesses over when packaging its fuel cell/battery. Always connect the battery **last**, disconnect **first**.

3. Reconciliation & Build Log

3.1 Step-by-Step Building Steps

1. **Chassis assembly** — lower FRP deck first: differential, rear T-bar/damper mount, front suspension towers. Torque suspension screws to spec but leave pivots free-moving.
2. **Drivetrain fitting** — install the brushless motor, mesh pinion-to-spur gear with a paper feeler gauge (slides out with light drag, doesn't bind or fall free), fit ESC with wiring clear of moving parts.
3. **Electronics wiring** — mount ESC/receiver away from differential grease and pinch points; set throttle/brake endpoints and LVC before the first battery connection.
4. **Steering linkage** — centre the servo electronically before attaching the steering rod, then attach with front wheels dead straight.
5. **Body/shell prep** — trim the Lexan body along marked lines, mask and paint from the **inside** (reverse order to a real car: clear areas masked first, base/metallic colour last against the inside surface).
6. **Livery application** — decals/masking after the base colour fully cures, same base-colour-first sequence a real F1 livery follows.
7. **Wing/aero trim fitting** — mount wings last, after the body is finished, so angle is set relative to actual ride height.
8. **Ride height & camber/toe setup** — set with the finished, painted body on the car, exactly as a real car is set up with its final aero package fitted.
9. **Shakedown run** — short, low-throttle first run confirming steering trim, motor rotation direction, brake feel.
10. **Tuning pass** — adjust damper oil weight, spring rate, camber/toe for the actual track surface — the same principle as a real team's practice-session setup work.

3.2 Engineering Trade-offs: Where Scale RC Physics Diverge from Real F1 Physics

- **Aerodynamics scale worse than almost anything else on the car.** Real downforce depends on velocity squared and a sealed, purpose-shaped underbody; shrinking the shape to 1:10 doesn't shrink the physics proportionally. RC cars run at a different Reynolds-number regime entirely, not just a smaller version of the full-size car's. Aero only becomes meaningfully useful above roughly a walking-to-jogging-pace-scaled threshold (research on small RC vehicles suggests ~22 mph), and even then the force is small relative to the car's weight compared to a full-size F1 floor.

- **RC underbodies are aerodynamically messy by default.** A stock RC chassis has an exposed, uneven underside (drivetrain, servo, ESC, screw heads) — most of the ground-effect story in §1.1 doesn't transfer without a dedicated, sealed scale floor most kits don't provide.
- **Tyres, not wings, dominate scale grip.** At RC speeds/weights, tyre compound and suspension setup contribute far more to cornering than any wing or floor detail — the inverse of a modern F1 car, where the floor is now the largest downforce contributor.
- **The power-unit story reverses.** Real F1 spends enormous effort harvesting/redeploying combustion-engine energy (MGU-K/H). An RC pan car is already 100% electric — there's no recovery problem, only a **heat and current-management** problem. A different electrical engineering challenge replaces the original one entirely.
- **Suspension geometry purpose diverges.** Real pushrod/pullrod geometry now partly protects ground-effect floor performance (anti-dive/anti-squat). An RC pan car's suspension is chosen almost entirely for mechanical grip and consistency — the same-looking linkage does a genuinely different job on each vehicle.
- **Weight-distribution logic is shared, but the "why" differs.** Both cars centralise and lower their biggest single mass. For the real car this also manages centre-of-gravity *change* as fuel burns off across a race. For the RC car, battery mass is fixed for the whole run — the benefit is purely the polar-moment-of-inertia/agility argument, the same design instinct arrived at from a narrower set of reasons.

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